Earth and Planetary Science Letters, 87 (1988) 453–462
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

[4]

Magnetostratigraphy of the Toarcian stratotype sections at Thouars and Airvault (Deux-Sèvres, France)

Bruno Galbrun [1], Jean Gabilly [2] and Léopold Rasplus [3]

[1] *Laboratoire de Stratigraphie, UA-CNRS 319, Université Paris VI, 4 place Jussieu, 75252 Paris Cedex (France)*
[2] *Laboratoire de Géologie Stratigraphique et Structurale, UA-CNRS 157, Université de Poitiers, 40 Avenue du Recteur Pineau, 86022 Poitiers Cedex (France)*
[3] *Laboratoire de Géologie, UA-CNRS 319, Université de Tours, Parc Grandmont, 37200 Tours (France)*

Received September 1, 1987; revised version accepted December 4, 1987

Magnetostratigraphic study of the Toarcian type sections of Thouars and Airvault (Deux-Sèvres, France) has yielded two reliable magnetic polarity sequences. Most samples were treated by mixed cleaning: thermal demagnetization (250°, 300° or 350° C) and subsequent alternating field demagnetization. Polarity intervals are easily identified and correlate well between the two sections using the biostratigraphic data provided by the detailed standard ammonite zonation of the Toarcian stage. The polarity sequence extends from ammonite horizon V (*Pseudoserpentinum* horizon, *Serpentinus* zone) to horizon XXV (*Subcompta* horizon, *Aalensis* zone); it shows 5 reversed and 5 normal polarity magnetozones.

1. Introduction

The magnetic polarity time scale for the Late Cretaceous and Cenozoic has been established by the correlation of marine magnetic anomalies with magnetostratigraphic sequences in sedimentary and volcanic rocks [1]. The correlation of the Late Jurassic to Early Cretaceous M-sequence anomalies with magnetostratigraphic studies in pelagic limestones sequences has been only partially established [2–4].

However, there is no oceanic crust known to be older than Middle Jurassic. Thus, the Early Jurassic magnetic polarity time scale cannot be derived from marine magnetic anomalies and must be established from magnetostratigraphic studies. Among sedimentary rock sequences, stratotype sections are the best documented ones with multiple biostratigraphic bed to bed analysis and, particularly during Jurassic times, with detailed ammonite zonation. The Toarcian stratotype was one of the chosen sections for two main reasons: (1) previous magnetostratigraphic studies which have investigated sedimentary rocks of Lower Jurassic age are scarce [5–7], (2) good correlation tools are needed for drillings projects, especially through the Toarcian series, well known as oil-source sedimentary facies in the French Paris Basin.

No radiometric data can be strictly attributed to the Toarcian stage, but it seems that this relatively long stage (8 ammonites zones) extends from 181 to 189 Ma [8].

We have sampled two sections in the southwest part of the Paris Basin: the Toarcian stratotype near Thouars and a complementary section in Airvault (20 km southeast) in order to build a magnetic polarity pattern. The results have yielded a magnetic polarity zonation and have provided very good biostratigraphic ages of the reversals based on an ammonite horizon scale. They have also located lacunae in the sedimentary record, better than with the ammonite zonation tool.

2. Geological setting, biostratigraphy and sampling

The Toarcian stratotype is located in the Paris Basin just southeast of the Palaeozoic outcrops of the Armorican Massif and north of the Poitou province. The sediments are clearly marine, formed in relatively shallow but quiet waters. Features of Toarcian coasts lay 5 km west where granitic coastal reliefs are known.

The Rigollier quarries in Vrines (47° N, 0.2° W), 2 km northwest of Thouars, have been designated as standard and first described by d'Orbigny [9] as "type-point" for the Toarcian stage. With a larger

0012-821X/88/$03.50 © 1988 Elsevier Science Publishers B.V.

TABLE 1

Ammonite zonation for the Toarcian in central-west France

Zones	Subzones	Horizons	N°
Aalensis	*Buckmani*	*Buckmani*	XXVII
	Aalensis	*Aalensis*	XXVI
	Mactra	*Subcompta*	XXV
		Mactra	XXIV
Pseudoradiosa	*Pseudoradiosa*	*Explanata*	XXIII
	Levesquei	*Levesquei*	XXII
Insigne	*Insigne*	*Gruneri*	XXI
		Insigne	XX
	Fallaciosum	*Fallaciosum*	XIX
Thouarsense	*Fascigerum*	*Fascigerum*	XVIII
	Thouarsense	*Thouarsense*	XVII
		Doerntense	XVI
	Bingmanni	*Bingmanni*	XV
Variabilis	*Vitiosa*	*Vitiosa*	XIV
	Illustris	*Illustris*	XIII
	Variabilis	*Variabilis*	XII
Bifrons	*Semipolitum*	*Semipolitum*	XI
	Bifrons	*Bifrons*	X
	Sublevisoni	*Lusitanicum*	IX
		Commune	VIII
		Sublevisoni	VII
Serpentinus	*Falciferum*	*Falciferum*	VI
		Pseudoserpentinum	V
	Strangewaysi	*Strangewaysi*	IV
		Elegantulum	III
Tenuicostatum	*Semicelatum*	*Semicelatum*	II
	Costatum	*Paltus*	I

view of the stratotype concept, we tend now to confer a regional setting to the Toarcian stratotype. At this regional scale, other sections such as the large "Ciments français" quarry at Airvault (46.8° N, 0.1° W) can also be chosen as reference.

The Toarcian stage was divided into 27 very well-defined ammonite horizons (Roman numerals in text, see Table 1) [10]. At Rigollier quarry (6.60 m thick), 25 of these horizons are found; horizons I and II are missing. The Airvault "Ciments français" quarry section (6.30 m thick) presents 23 regional horizons: IV to XXVII.

The horizons belong to two lithological units (see lithologic columns in Figs. 6 and 7).

Lower unit. At Thouars it begins by horizon III with reddish-brown, lenticular sandstones, overlying eroded Pliensbachian sandstones. Horizon IV is very thin: only a few centimeters of grey sandy limestone. Horizons V to XII are represented through 1 m reddish-brown bioclastic iron-oolith limestone.

At Airvault, the basal bed (horizon IV) is a thin (0.15 m), lenticular, grey, sandy limestone bed, overlying a very reduced Pliensbachian or in some places directly the Brioverian basement. As in Thouars, horizons V to XII are formed by a slightly thicker (1.35 m) prominent reddish-brown, bioclastic iron-oolith limestone member where ammonite horizons are more distinct. This limestone bar is regionally typical at the top of the lower unit.

Upper unit. A lithological change occurs above horizon XII. At Thouars, horizons XIII to XXVII are interbedded dark grey marls and centimetre thick layers of grey, clayey and micritic limestone. End of horizon XVIII and horizons XIX and XX are more resistant clayey micritic limestones prominent on the quarry working face. However, iron-ooliths persist only up to the top of horizon XXV.

At Airvault, this unit consists of dark grey marls and limestone without iron-oolith. Horizons XVII and XVIII (0.25 m) are more condensed than at Thouars. The top of horizon XVIII is oxydized and bored, and corresponds to a disconformity. Horizons XIX, XX and XXI are very condensed. A conglomeratic level rich in reworked, bored phosphatic pebbles and nodules occurs at the base of horizon XIX.

In both sections the sedimentation rate increases in the upper part of this unit.

Because the strata are horizontal, they can be followed in exposures and sampled without any problem. Every limestone bed as well as some interbedded marl layers were sampled in both sections. The mean sampling interval was about 0.1 m.

3. Magnetic properties

3.1. Isothermal remanent magnetization (IRM) and susceptibility

Coercivity spectrum analysis by acquisition of isothermal remanent magnetization (IRM) and subsequent thermal demagnetization, as an effec-

tive means of characterizing the magnetic carriers of marine limestones [11], has been applied to a set of samples representative of each lithology, marly and iron-oolith limestones. The IRM was given in progressive DC fields up to 1.6 T using an electromagnet. Typical results are illustrated in Fig. 1.

IRM acquisition curves of marly limestone samples are characterized by saturation in fields as low as 0.2 T (Fig. 1a). Thermal demagnetization curves indicate a maximum blocking temperature of 570 °C. These characteristics show that magnetite is the main carrier of magnetization in these marly limestones.

The IRM of iron-oolith limestones is characterized by a broad range of coercivities (Fig. 1b), saturation being unattained at 1.6 T. IRM increases rapidly up to 0.1 T and more slowly up to 1.6 T. Progressive thermal demagnetization of the maximum IRM shows a rapid decrease of the magnetization below 60 °C, which is characteristic of goethite [11]. Above this temperature, the decrease of remaining IRM up to a maximum blocking temperature of 570 °C indicates that magnetite is also present in the iron-oolith limestones.

In both cases thermal demagnetization shows a small persistent component above 600 °C. This high blocking temperature mineral is probably hematite and may result from alteration of the magnetic minerals at high temperature.

Magnetite is the main ferrimagnetic phase of the limestones from Thouars and Airvault, and it

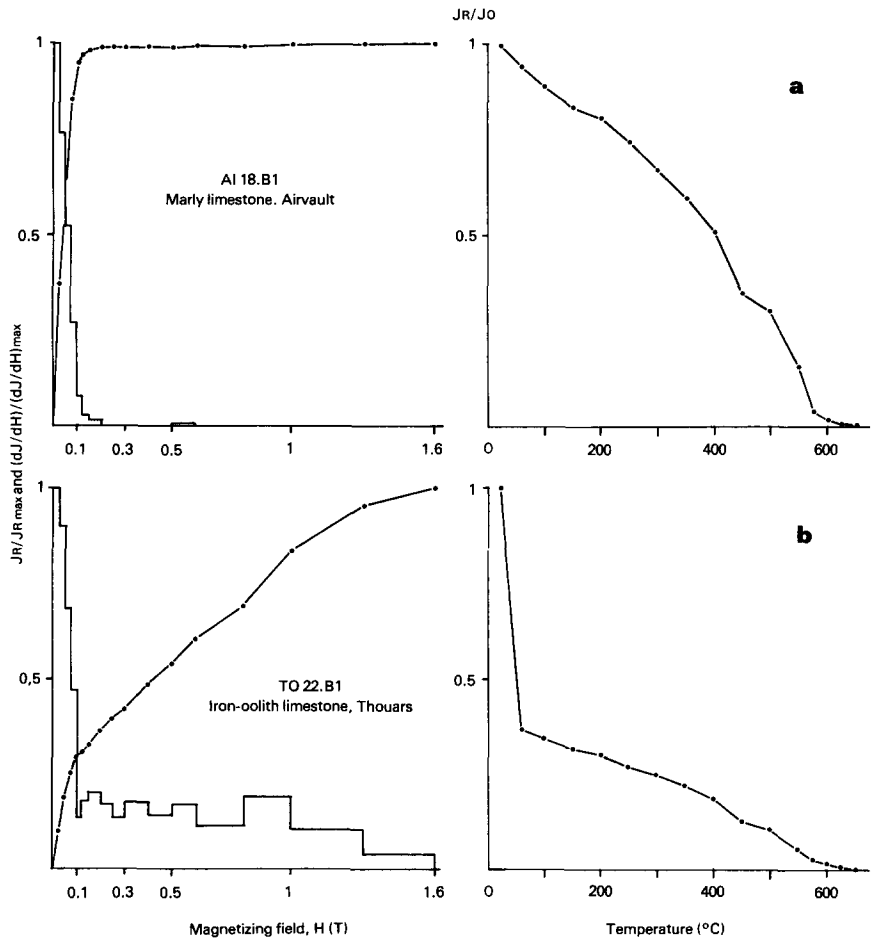


Fig. 1. Acquisition of isothermal remanent magnetization (IRM) and subsequent thermal demagnetization of the maximum IRM. See explanation in text.

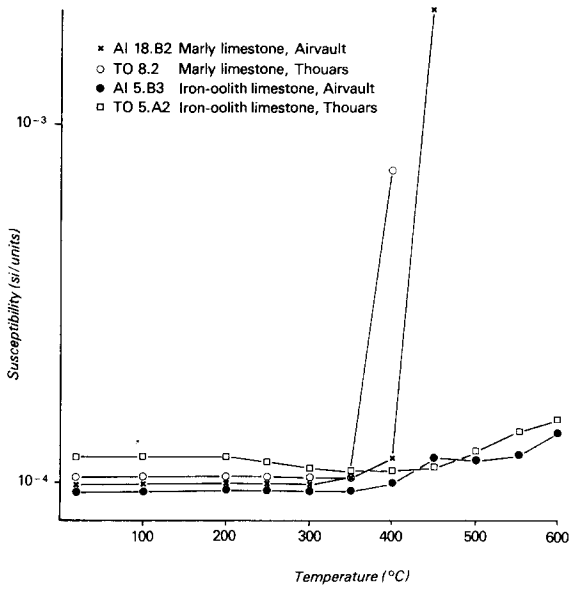


Fig. 2. Susceptibility variation in samples of marly and iron-oolith limestones after heating to the indicated temperature. The sharp increase of susceptibility after heating to 350–400 ° C indicates growth of a new magnetic phase in marly limestones.

is probable that a detrital remanent magnetization can be isolated. Goethite probably originates from the alteration of ferruginous ooliths.

Magnetic alterations that can occur in marine limestones during thermal cleaning, may be detected by susceptibility measurements after each heating stage. The susceptibility of some samples was monitored after heating using a KLY-2 low-field susceptibility bridge (Geofyzika Brno) (Fig. 2). In the iron-oolith limestones no significant increase of susceptibility until 600 ° C was observed, thus indicating apparently that no magnetic alteration occurred during thermal demagnetization. However, the susceptibility of the marly limestone shows a sharp increase after heating to 350–400 ° C. This indicates the growth of a new magnetic phase with high susceptibility, probably magnetite. The change may be attributed to the destruction of clay minerals. Thus for samples of marly limestone thermal treatment must be limited to 350 ° C.

3.2. Measurement and analysis of the natural remanent magnetization (NRM)

The remanent magnetization of the samples was measured using a three-axis RS-01 cryogenic magnetometer (LETI-CEA). Remanences weaker than 2.5×10^{-5} A/m could not be measured with good precision and were rejected.

The natural remanent magnetization (NRM) intensities range from 0.05 to 2×10^{-3} A/m in the Thouars section and 0.03 to 1.2×10^{-3} A/m in the Airvault section. At the top of both sections the NRM intensity drops below 10^{-4} A/m. At Thouars this decrease corresponds to the disappearance of ferruginous ooliths near the top of the *Subcompta* ammonite horizon. At Airvault this decrease occurs in the middle of the *Explanata* horizon, but does not correspond to a lithological change.

All NRM directions in both sections were near the present geomagnetic field and were entirely of normal polarity (Fig. 3). The results of magnetite cleaning were analyzed using the classical method of linear segments in orthogonal vector plots [12]. Fig. 4 shows some typical demagnetization plots.

Thermal demagnetization at 250–300 ° C removes an initial unstable component and delineates a vector which decays to the origin of the vector diagram above 300 ° C (Fig. 4a). The direction of this vector determines the characteristic remanent magnetization (ChRM). But some magnetic alterations (erratic changes in direction and intensity of NRM, strong viscous behavior above

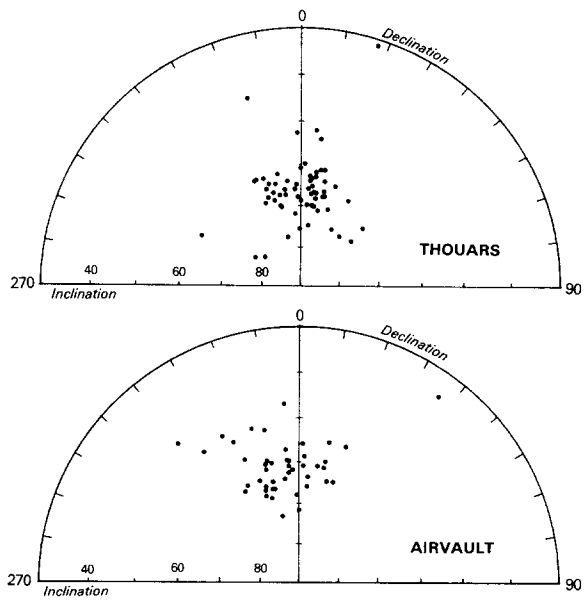


Fig. 3. Equal area stereographic plots of the NRM directions.

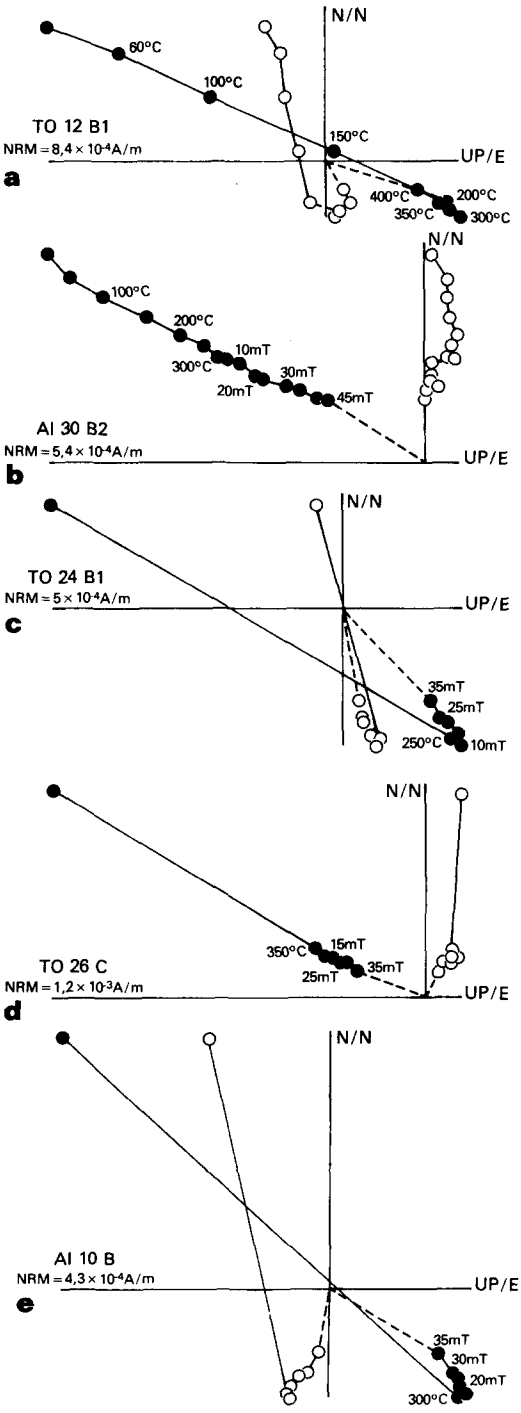


Fig. 4. Orthogonal projections of thermal demagnetization (a) and mixed (thermal and subsequent alternating field) demagnetization (b–e). An initial unstable component is removed by 250–300 °C heating. Subsequent thermal or alternating field demagnetizations define the characteristic remanent magnetization (ChRM).

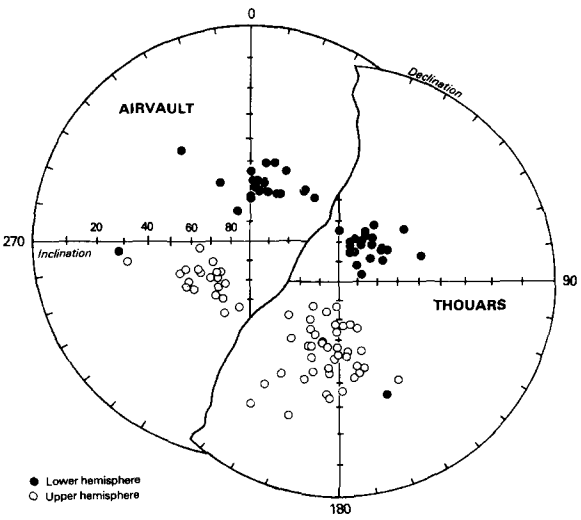


Fig. 5. Equal area stereographic plots of the characteristic remanent magnetization directions.

400 °C) occur at high temperature in most samples, especially in marly limestones (cf. susceptibility measurement after heating in Fig. 2). Thus, thermal cleaning (250 °, 300 ° or 350 °C) and subsequent alternating field demagnetization were used for most samples (Fig. 4b–e). This mixed cleaning technique, classical in paleomagnetic studies [13], effectively removed the unstable component of normal polarity. The ChRM directions are defined above 20 mT and are of normal (Fig. 4b, d) or reversed polarity (Fig. 4a, c, e). Very weak NRM intensities prevented the isolation of a stable component in samples from the upper part of both sections.

Seventy-eight samples from the Thouars section were studied, of which 65 yielded reliable results; from Airvault 70 were studied, stable directions being obtained from 44 of them.

TABLE 2

Characteristic magnetization mean directions for the Thouars and Airvault sections

Section	Samples	N	D (°)	I (°)	α_{95} (°)	k
Thouars	normal	21	38.4	66.8	3.7	68.5
	reversed	40	191	−53.1	4.7	22.5
	combined	61	18.1	58.5	3.8	21.6
Airvault	normal	20	15.9	59	5.6	31.8
	reversed	19	226.2	−60.1	4.4	54.1
	combined	39	30.5	60.4	4.1	30

The ChRM vectors form nearly antipodal well clustered groups of normally and reversely magnetized samples (Fig. 5). The mean directions of characteristic magnetization are listed in Table 2; detailed analysis of paleomagnetic results has already been published [14].

Considering the magnetic characteristics of the samples, it could be assumed that the unstable component is probably a viscous remanent magnetization and the stable component carried by magnetite was acquired shortly after deposition.

4. Magnetostratigraphy

The stratigraphic distribution of ChRM directions gives two excellent magnetic polarity sequences which are presented in Figs. 6 and 7 for the Thouars and Airvault sections respectively.

At Thouars, horizons III, IV and V (*Elegantulum, Strangewaysi* and *Pseudoserpentinum* horizons) did not give stable paleodirections. Thus, the magnetic polarity sequence begins with horizons VI, VII, VIII and IX (*Falciferum, Sublevisoni, Commune* and *Lusitanicum* horizons) through a single iron-oolith limestone layer of normal polarity. It ends with the top of the *Subcompta* reversed polarity horizon (XXV). Five normal polarity zones and five reversed polarity zones, each defined by several samples, constitute the sequence. The marly *Illustris* horizon (XIII) alone could not be sampled, so the sequence is quite complete, except for a 0.15 m hiatus. But this horizon is partly sampled in the Airvault section.

At Airvault, the polarity sequence begins with horizon V and extends up to the middle of horizon

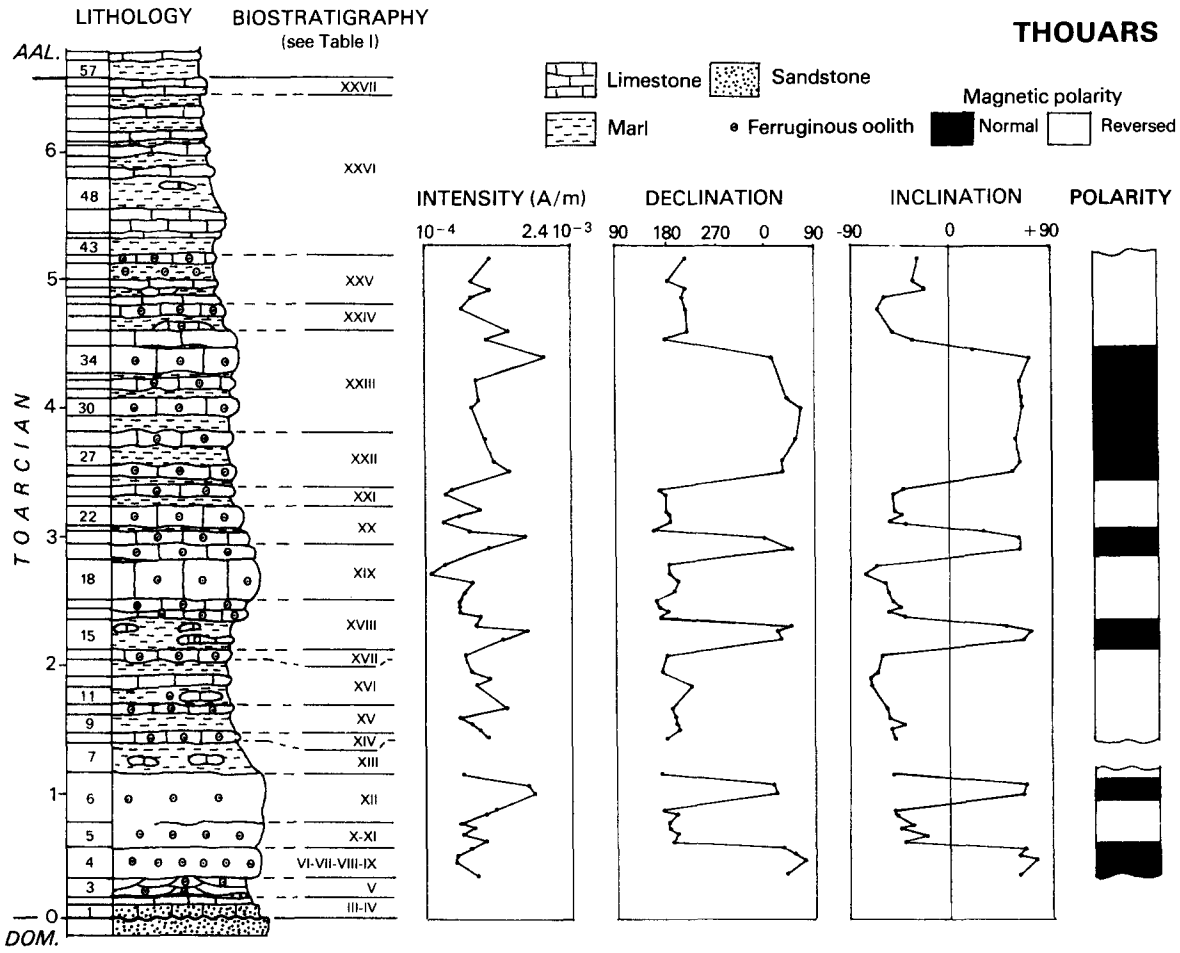


Fig. 6. Lithostratigraphy, biostratigraphy and magnetostratigraphy of the Thouars section.

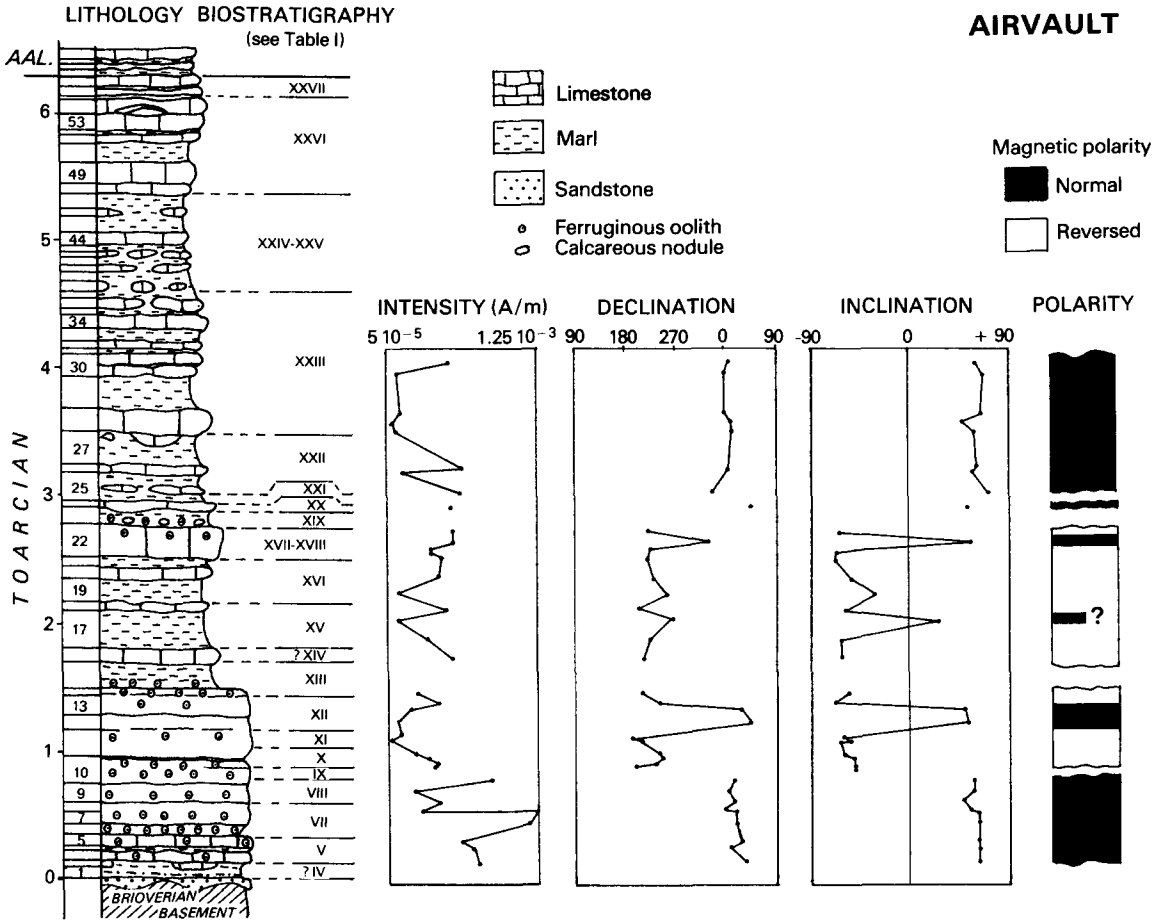


Fig. 7. Lithostratigraphy, biostratigraphy and magnetostratigraphy of the Airvault section.

XXIII (*Explanata* horizon). Horizons IX, XIX, XXI (*Lusitanicum, Fallaciosum* and *Gruneri* horizons) could not be sampled so the polarity zonation is less complete than in Thouars. But all these missing horizons have been studied in Thouars.

The magnetic polarity patterns of the two sections can be relatively well correlated but there are some slight differences (Fig. 8). First, the base of the *Variabilis* horizon (XII) is reversed in Thouars while it is of normal polarity in Airvault. This fact suggests a brief hiatus at the base of the condensed horizon XII at Airvault. Second, in the upper part of the *Bigmanni* horizon (XV), one of the samples from the Airvault section gave a doubtful normal polarity. This normal polarity zone does not appear in Thouars. Third, the horizons XVII and XVIII (*Thouarsense* and *Fascigerum* horizons) are represented by several layers at Thouars and by a single bed at Airvault. This biostratigraphic interval contains two reversed polarity zones separated by a normal one, in both sections. The normal interval is documented by a single sample at Airvault and several at Thouars, where normal polarity appears at the base of horizon XVIII. Thus, the boundary between horizons XVII and XVIII can be precisely positioned at Airvault, in a single bed, by magnetostratigraphic means. Fourth and last, in Airvault, horizons XIX to XXI are condensed and a single limestone bed representing horizon XX (*Insigne* horizon) could be sampled and normal polarity was found. At Thouars, horizon XX is a two-bed horizon. The polarity is normal for the lower bed and reversed for the upper bed. This last bed is missing at Airvault as well as the thin underlying marly interval which separates the two beds at

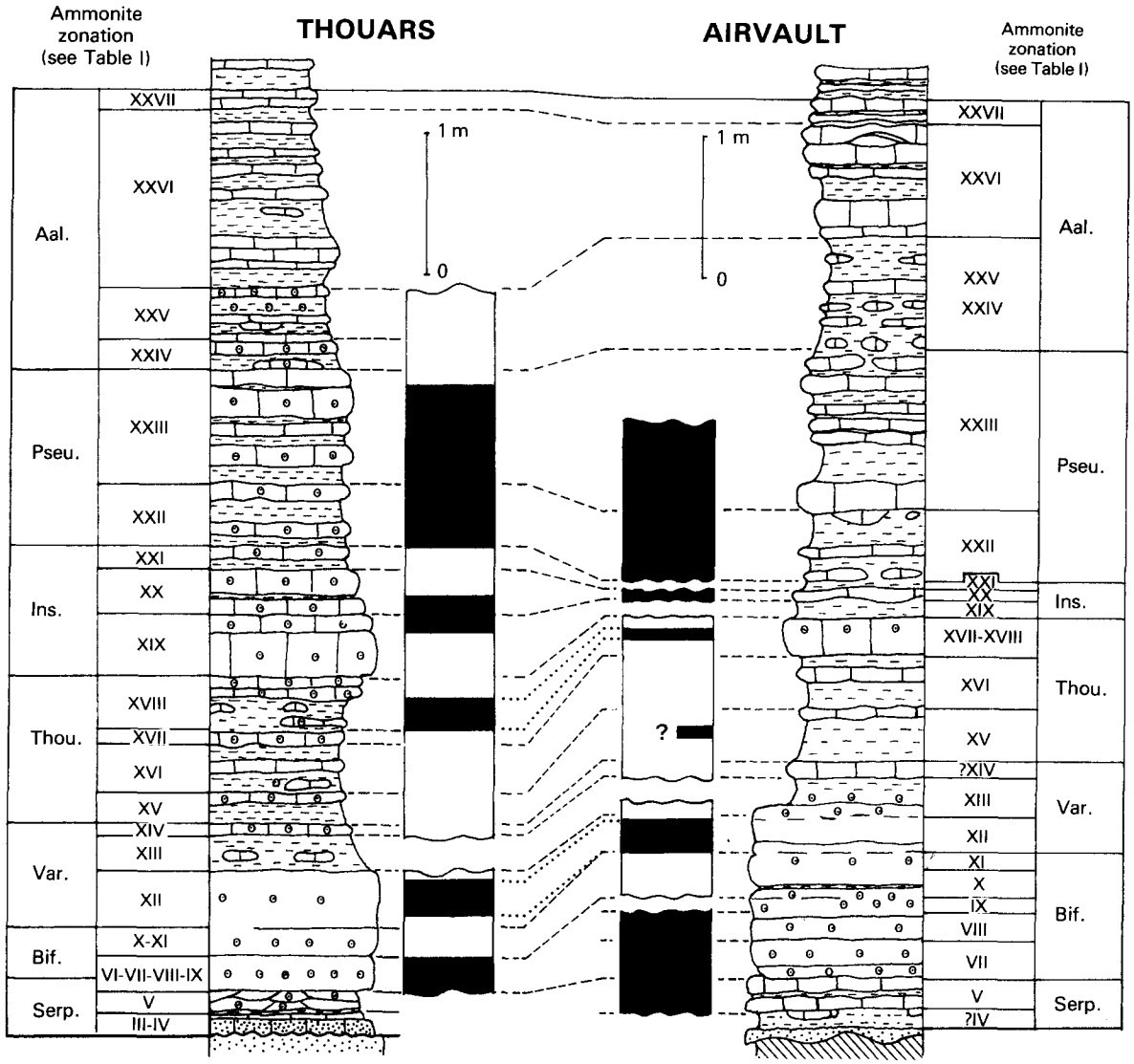


Fig. 8. Biostratigraphic and magnetostratigraphic correlation between the two sections. Black is normal polarity, white is reversed polarity. Solid tie lines are ammonite horizon boundaries; dotted lines are the complementary correlations of polarity intervals.

Thouars. The reversed younger interval is still represented by the multiple sampled horizon XXI at Thouars while at Airvault, the soft marly single layered horizon XXI could not be sampled. Thus, at Airvault the younger reversed interval (which overlaps the upper part of horizon XX and the whole horizon XXI at Thouars) has yet to be identified.

Finally, the composite magnetic column is fairly well defined by the two sections and slight distorsions are the result of modified sedimentation processes or sampling impossibility. This Toarcian magnetic column must be completed with other outcrops showing the lowest ammonite *Tenuicostatum* zone (horizons I and II), the *Strangewaysi* subzone (horizons III and IV) and the upper part of the upper Toarcian, i.e., *Aalensis* (XXVI) and *Buckmanni* (XXVII) subzones/horizons.

5. Discussion and conclusion

The polarity sequence obtained from the very well dated Toarcian stratotype section of Thouars

completed by a section at Airvault is fairly well defined. It represents a good approach of the magnetic polarity history during the main part of Toarcian time. It extends from ammonite horizon V (*Pseudoserpentinum* horizon, *Serpentinus* zone) to top of horizon XXV (*Subcompta* horizon, *Aalensis* zone). The magnetostratigraphic zonation agrees well with the biostratigraphic and lithologic correlations.

This is the first Jurassic magnetic polarity sequence dated with a precision of one ammonite horizon. This is particularly important during the Liassic period for which no oceanic anomalies sequence is available. In this case, it may be a very useful tool for core interpretations and for correlations between West European, Boreal and Tethyan fauna realms.

Few magnetostratigraphic studies are available for Toarcian time in the Tethyan realm [6,7]. They gave given different magnetic polarity sequences which also seem different from the stratotype sequence. Correlations between Tethyan and West European ammonite zonations are not easy: the lower Toarcian zonations seem the same in both realms, but the Middle Toarcian Tethyan zonation presents the *Bifrons* zone with two subzones (*Sublevisoni* and *Semipolitum* subzones) while in West European zonation the *Bifrons* zone is divided into three subzones (*Sublevisoni, Bifrons* and *Semipolitum* subzones). The *Semipolitum* subzone basis is probably the same stratigraphic level in both zonations. Between the Tethyan Upper Toarcian (*Erbaense* and *Meneghinii* zones) and West European zonations (*Thouarsense, Insigne, Pseudoradiosa* and *Aalensis* zones), the *Meneghinii* zone base can be correlated with the *Pseudoradiosa* zone base. With these poor biostratigraphic correlations, the various polarity patterns cannot be compared because the stratotype zonation is more precise than in other sections, strong facies changes occur, and sampling scales are different.

As the Toarcian magnetic sequence established in this study does not cover the whole Toarcian stage (lacunae at the lower and the upper parts of the stratotype sections), it is difficult to appreciate the mean duration of a magnetozone. However, the average frequency of reversals seems to be 2–3 reversals per ammonite zone and is smaller than in the Pliensbachian (about 8–10 reversals per ammonite zone) [6].

Acknowledgements

We thank M. Cremer and officials of the cement works "Ciments français" (Airvault) for permission to sample in their quarry. Special thanks are extended to Dr. L. Daly who made available the facilities of Laboratoire de Géomagnétisme (Université Paris VI) for the IRM and susceptibility measurements. Prof. J. Dercourt and Dr. L. Daly read and criticized the manuscript. We appreciate reviews provided by Dr. M. Westphal and two anonymous reviewers. We are grateful to John Thompson for revision of the English.

References

1 W. Lowrie and W. Alvarez, One hundred million years of geomagnetic polarity history, Geology 9, 392–397, 1981.
2 M.B. Steiner, J.G. Ogg, G. Melendez and L. Sequeiros, Jurassic magnetostratigraphy, 2. Middle-Late Oxfordian of Aguilon, Iberian Cordillera, northern Spain, Earth Planet. Sci. Lett. 76, 151–166, 1985.
3 B. Galbrun, La séquence de polarité magnétique au passage Jurassique-Crétacé: corrélations entre les données magnétobiostratigraphiques et la succession des anomalies magnétiques océaniques, C.R. Acad. Sci. Paris 303, 495–498, 1986.
4 W. Lowrie and J.G. Ogg, A magnetic polarity time scale for the Early Cretaceous and Late Jurassic, Earth Planet Sci. Lett. 76, 341–349, 1986.
5 E. Marton, P. Marton and F. Heller, Remanent magnetization of a Pliensbachian limestone sequence at Bakonycsernye (Hungary), Earth Planet. Sci. Lett. 48, 218–226, 1980.
6 F. Horner and F. Heller, Lower Jurassic magnetostratigraphy at the Breggia Gorge (Ticino, Switzerland) and Alpe Turati (Como, Italy), Geophys. J.R. Astron. Soc. 73, 705–718, 1983.
7 J.E.T. Channell, W. Lowrie, P. Pialli and F. Venturi, Jurassic magnetic stratigraphy from Umbrian (Italian) land sections, Earth Planet. Sci. Lett. 68, 309–325, 1984.
8 W.J. Kennedy and G.S. Odin, The Jurassic and Cretaceous time scale in 1981, in: Numerical Dating in Stratigraphy, G.S. Odin ed., pp. 557–592, John Wiley and Sons, New York, N.Y., 1982.
9 A. d'Orbigny, Cours élémentaire de paléontologie et de géologie stratigraphiques, 847 pp., Masson, Paris, 1849.
10 J. Gabilly, Le Toarcien à Thouars et dans le Centre-Ouest de la France, biostratigraphie, évolution de la faune (*Harpoceratinae, Hildoceratinae*). Les stratotypes français, 3, 217 pp., CNRS, Paris, 1976.
11 W. Lowrie and F. Heller, Magnetic properties of marines limestone, Rev. Geophys. Space Phys. 20, 171–192, 1982.
12 J.D.A. Zijderveld, A.C. demagnetization of rocks: analysis of results, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn eds., pp. 254–286, Elsevier, Amsterdam, 1967.

13 L. Daly, Des aimantations partielles aux aimantations superposées: espoirs et difficultés, Phys. Earth Planet. Inter. 24, 218–227, 1981.
14 B. Galbrun and L. Daly, Etablissement d'un pôle paléomagnétique pour la plate-forme européenne stable à l'époque toarcienne, C.R. Acad. Sci. Paris 305, 773–776, 1987.